Realised Volatility Forecasting — Multi-Asset Benchmark

Direct Multi-Step Models: HAR, NLinear, DLinear, SAS, RNN

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1 Setup

$x_t = \log \mathrm{RV}_t$, z-scored per rolling window (μ, σ frozen until refit). Direct multi-step: one model per $h \in \{1, 5, 10\}$. Ridge $\alpha \in \{10^k : k \in [-4, 5],\ \Delta k{=}0.25\}$ (37 values), 5-fold CV.

2 Models

2.1 HAR — Heterogeneous Autoregressive Model (Corsi 2009)

Features. Let $\bar{x}_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} x_{t-k}$ and $\bar{x}_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} x_{t-k}$.

$$\hat{x}_{t+h} = \alpha + \beta_d\, x_t + \beta_w\, \bar{x}_t^{(w)} + \beta_m\, \bar{x}_t^{(m)} \tag{1}$$

HAR-OLS: $\min_\beta \|X\beta - y\|_2^2$. **HAR-Ridge**: column-normalise $[x^{(d)}, x^{(w)}, x^{(m)}]$; penalise slopes only; back-transform $w = w_{\text{scaled}}/\sigma_{\text{col}}$.

2.2 NLinear and DLinear (Zeng et al. 2023)

Let $\boldsymbol{x}_t = (x_{t-L+1}, \ldots, x_t)^\top \in \mathbb{R}^L$ be the lookback window.

NLinear.

$$\tilde{\boldsymbol{x}}_t = \boldsymbol{x}_t - x_t, \qquad \hat{x}_{t+h} = \tilde{\boldsymbol{x}}_t^\top w_h + x_t \tag{2}$$

DLinear.

$$\boldsymbol{\tau}_t = \mathrm{MA}_k(\boldsymbol{x}_t), \quad \boldsymbol{\varsigma}_t = \boldsymbol{x}_t - \boldsymbol{\tau}_t, \qquad \hat{x}_{t+h} = \begin{bmatrix} \boldsymbol{\tau}_t \\ \boldsymbol{\varsigma}_t \end{bmatrix}^\top w_h \tag{3}$$

2.3 GARCH — AR(p)-GARCH(1,1)

$x_t = \mu + \sum_{i=1}^{p} \phi_i x_{t-i} + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G \sigma_{t-1}^2$. Multi-step: iterate variance recursion analytically. QMLE.

2.4 SAS — State Affine System

Diagonal basis. $P_0, \ldots, P_p \in \mathbb{R}^n,\ Q_0, \ldots, Q_q \in \mathbb{R}^n$.

$$P(z) = \mathrm{clip}\Big(\sum_{d=0}^{p} P_d\, z^d,\ -1{+}\epsilon,\ 1{-}\epsilon\Big) \in (-1,1)^n \quad \text{[Horner's rule]} \tag{4}$$

$$Q(z) = \sum_{d=0}^{q} Q_d\, z^d \in \mathbb{R}^n \tag{5}$$

$$s_t = P(z_t) \odot s_{t-1} + Q(z_t) \tag{6}$$

Clipping ensures $|P(z)_i| < 1$ (unconditional stability). Readout: $\hat{x}_{t+h} = s_t^\top w_h$.

Associative structure. $\ell_t = (P(z_t), Q(z_t))$. Composition rule:

$$(A_j,\, b_j) \circ (A_i,\, b_i) \;=\; \big(A_j \odot A_i,\;\; A_j \odot b_i + b_j\big) \tag{7}$$

is associative $\Rightarrow$ parallel divide-and-conquer scan.

Algorithm 1: Two-Level Parallel Scan — compute $s_1, \ldots, s_T$ from $z_1, \ldots, z_T$

Input : Transitions $\ell_1, \ldots, \ell_T$;;
chunk size B;;
$s_0 = \mathbf{0}$
Output : States $s_1, \ldots, s_T$;;
s_T

$K \leftarrow \lceil T/B \rceil$; pad to KB
Phase 1 (vmap over K chunks);
  $\forall\, k:\ (A^{\text{intra}}_{k,\cdot},\, b^{\text{intra}}_{k,\cdot}) \leftarrow \textsc{AssocScan}(\circ,\ \ell_{kB+1}, \ldots, \ell_{(k+1)B})$
Phase 2 (inter-chunk scan over K endpoints);
  $(A^{\text{inter}},\, b^{\text{inter}}) \leftarrow \textsc{AssocScan}(\circ,\ (A^{\text{intra}}_{1,B}, b^{\text{intra}}_{1,B}), \ldots)$
Phase 3 (carry for chunk $k \geq 1$);
  $c_0 = s_0$;;
$c_k = A^{\text{inter}}_{k-1} \odot s_0 + b^{\text{inter}}_{k-1}$
Phase 4 (vmap over K chunks);
  $\forall\, k, j:\ s_{kB+j} = A^{\text{intra}}_{k,j} \odot c_k + b^{\text{intra}}_{k,j}$

Depth $O(\log T)$ vs. $O(T)$ sequential.

Augmented SAS (AugSAS).

$$f_t = \big[s_t;\ 1,\ x_t,\ \bar{x}_t^{(w)},\ \bar{x}_t^{(m)}\big] \in \mathbb{R}^{n+4}, \qquad \hat{x}_{t+h} = f_t^\top w_h \tag{8}$$

$w_h[1{:}n] = \mathbf{0}$ recovers HAR $\Rightarrow$ AugSAS $\geq$ HAR by construction.

2.5 MdSAS — Multi-Degree State Affine System

Motivation. In a single SAS reservoir every state dimension uses the same P-polynomial degree. A degree-0 filter ($P = P_0$) is a pure exponential smoother — essentially an AR(1) memory. A degree-1 filter ($P = P_0 + P_1 z$) multiplies the memory decay by the current observation, creating an input-gated, nonlinear interaction. These two sub-systems may require very different ridge penalties: the linear memory is globally stable while the gating term creates local heteroscedasticity. Assigning them a single shared α forces a compromise that can under-regularise one group or over-regularise the other.

Architecture. For $D+1$ groups ($d = 0, \ldots, D$), create independent sub-reservoirs each of size n:

$$s_t^{(d)} = P^{(d)}(z_t) \odot s_{t-1}^{(d)} + Q^{(d)}(z_t), \qquad d = 0, \ldots, D. \tag{9}$$

The concatenated state $\tilde{s}_t = [s_t^{(0)}; \ldots; s_t^{(D)}] \in \mathbb{R}^{n(D+1)}$ is the feature vector.

Block-diagonal ridge. The readout solves a ridge regression with a *block-diagonal* penalty matrix $\Lambda = \mathrm{diag}(\alpha_0 \mathbf{I}_n, \ldots, \alpha_D \mathbf{I}_n)$, i.e. one scalar α_d per group. Given the precomputed Gram matrix $S^\top S$, iterating over the Cartesian product of per-group grids and modifying only the diagonal incurs negligible overhead.

Ablation variants.

- $\mathbf{SAS_{uniform}}$: standard single reservoir, p=1, single α. Baseline.
- $\mathbf{MdSAS_{arch}}$: two sub-reservoirs (p_0=0, p_1=1), *shared* α. Isolates the architectural benefit of mixing degrees from the regularisation benefit.
- $\mathbf{MdSAS_{d1}}$: same architecture, *per-group* α. Adds independent regularisation on top.
- $\mathbf{MdSAS_{d2}}$: three sub-reservoirs (p_0=0, p_1=1, p_2=2), per-group α. Tests the value of a degree-2 extension.

2.6 RNN — LSTM / GRU / LRU

Readout: $\hat{x}_{t+h} = \mathrm{Linear}_h(h_T)$, one head per h. Training: Adam, early stopping (patience=20, val=20% chronological).

LRU (Orvieto et al. 2023):

$$h_t = \sigma(\nu) \odot h_{t-1} + Bz_t, \qquad y_t = Ch_t, \qquad \sigma(\nu) \in (0,1)^d \tag{10}$$

3 Evaluation Protocol

Algorithm 2: Rolling Out-of-Sample Evaluation

Input : Series $x_1, \ldots, x_T$; window W; refit frequency R; horizons $\mathcal{H}$
Output : Squared-error sequences $\{e_{t,m}^{(h)}\}$ for each model m, horizon h

```
(μ, σ) ← (0, 1);
for t = W to T − max(H) do
    train ← x_{t−W:t};
    if t mod R = 0 then
        (μ, σ) ← FitScaler(train);
        traiñ ← (train − μ)/σ;
        foreach model m do
            m.Fit(traiñ, H)
    z_t ← (x_t − μ)/σ;
    foreach model m, horizon h ∈ H do
        ẑ_{t+h} ← m.Predict(h);
        e^{(h)}_{t,m} ← (ẑ_{t+h} − z_{t+h})^2;
    foreach model m do
        m.Update(z_t)
```

Model Confidence Set (Hansen et al. 2011). $L \in \mathbb{R}^{T \times M}$, row-normalised by cross-model mean to prevent high-vol dominance.

1. $d_{t,m} = L_{t,m} - \bar{L}_t$
2. $T_R = \max_m \bar{d}_m / \hat{\sigma}_m$ ($\hat{\sigma}_m$: circular block bootstrap, B reps)
3. Eliminate worst model if $p < \alpha$; repeat until H_0 not rejected.

4. Surviving set = MCS_α.

MCS Frequency. Run MCS per symbol per horizon; report $c_m^{(h)}/N$.

4 Hyperparameters

Table 1: Fixed hyperparameters used in all experiments.

Component	**Parameter**	**Value**
Evaluation	Training window W	2 000
	Refit frequency R	20 steps (≈1 month)
	Horizons $\mathcal{H}$	$\{1, 5, 10\}$ days
MCS	Significance α	0.10
	Bootstrap replications B	1 000
HAR	Lags (d, w, m)	$(1, 5, 22)$ days
	Ridge CV folds	5
NLinear / DLinear	Lookback L	20
	DLinear MA kernel	5
GARCH	AR lags p	5
	Estimation	QMLE
SAS (diagonal)	Reservoir size n	100 or 200
	Poly degree (p, q)	$(1, 1)$ or $(2, 2)$
	Spectral norm constraint	{0.80, 0.90, 0.95, 0.99}
	Washout	50 steps
RNN (GRU)	Hidden size	32
	Layers	1
	Lookback	20
	Epochs (max)	500
	Early stopping	patience = 20, val = 20%
MdSAS	Groups (D+1)	2 (degrees 0,1) or 3 (degrees 0,1,2)
	n per group	50 or 100
	Spectral norm ρ	{0.90, 0.95}
	α grid (grouped)	7-value subset (Δk=1.5, 49 combos)
Ridge grid	Range	10^k, $k \in [-4, 5]$
	Spacing (single-α)	$\Delta k = 0.25$, 37 values
	Spacing (grouped)	$\Delta k = 1.5$, 7 values (49 combos/2 groups)

5 Results

5.1 HAR vs. SAS (q=1, q=2) — 30 Symbols, $h \in \{1, 5, 10\}$

Table 2: Out-of-sample MSE — mean ± std across 30 symbols (W=2000, R=20, target: $\log \mathrm{RV}_5$, N_{sym}=30). Bold: lowest mean per horizon. $\mathrm{SAS}_{\mathrm{q1}}$: n=200, sn=0.95, q=1. $\mathrm{SAS}_{\mathrm{q2}}$: same but q=2 (quadratic input drive).

Model	$h = 1$	$h = 5$	$h = 10$	avg
HAR	0.400 ± 0.072	0.509 ± 0.093	0.583 ± 0.105	0.497
$\mathrm{SAS}_{\mathrm{q1}}$	0.416 ± 0.095	0.533 ± 0.196	0.730 ± 0.861	0.560
$\mathrm{SAS}_{\mathrm{q2}}$	$\mathbf{0.396 \pm 0.072}$	$\mathbf{0.507 \pm 0.094}$	$\mathbf{0.586 \pm 0.114}$	**0.496**

Table 3: Model Confidence Set (MCS, α=0.10, B=1000 bootstrap replications, Hansen et al. 2011) — frequency of inclusion in the superior set across 30 symbols. A model "survives" the MCS if it cannot be shown to be significantly worse than the best model on that symbol.

Model	$h = 1$	$h = 5$	$h = 10$	total
HAR	7/30 (23%)	13/30 (43%)	13/30 (43%)	33/90
$\mathrm{SAS}_{\mathrm{q1}}$	28/30 (93%)	30/30 (100%)	30/30 (100%)	88/90
$\mathrm{SAS}_{\mathrm{q2}}$	23/30 (77%)	24/30 (80%)	20/30 (67%)	67/90